

[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 1132

C

Unique Paper Code : 32357501

Name of the Paper : DSE-I Numerical Analysis
(LOCF)

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : V

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. All **six** questions are compulsory.
3. Attempt any two parts from each question.
4. Use of non-programmable scientific calculator is allowed.

P.T.O.

1. (a) Define fixed point of a function and construct an algorithm to implement the fixed point iteration scheme to find a fixed point of a function. Find the fixed point of $f(x) = 2x(1 - x)$. (6)
- (b) Perform four iterations of Newton's Raphson method to find the positive square root of 18. Take initial approximation $x_0 = 4$. (6)
- (c) Find the root of the equation $x^3 - 2x - 6 = 0$ in the interval (2, 3) by the method of false position. Perform three iterations. (6)
2. (a) Define the order of convergence of an iterative method for finding an approximation to the root of $g(x) = 0$. Find the order of convergence of Newton's iterative formula. (6.5)
- (b) Find a root of the equation $x^3 - 4x - 8 = 0$ in the interval (2, 3) using the Bisection method till fourth iteration. (6.5)

- (c) Perform three iterations of secant method to determine the location of the approximate root of the equation $x^3 + x^2 - 3x - 3 = 0$ on the interval $(1, 2)$. Given the exact value of the root is $x = \sqrt{3}$, compute the absolute error in the approximations just obtained. (6.5)

3. (a) Using scaled partial pivoting during the factor step, find matrices L , U and P such that $LU = PA$

where $A = \begin{pmatrix} 1 & 1 & 2 \\ -1 & 0 & 2 \\ 3 & 2 & -1 \end{pmatrix}$ (6.5)

- (b) Set up the SOR method with $w=0.7$ to solve the system of equations:

$$3x_1 - x_2 + x_3 = 4$$

$$2x_1 - 6x_2 + 3x_3 = -13$$

P.T.O.

$$-9x_1 + 7x_2 - 20x_3 = 7$$

Take the initial approximation as $X^{(0)} = (0, 0, 0)$
and do three iterations. (6.5)

- (c) Set up the Gauss-Jacobi iteration scheme to solve the system of equations:

$$10x_1 + x_2 + 4x_3 = 31$$

$$x_1 + 10x_2 - 5x_3 = -23$$

$$3x_1 - 2x_2 + 10x_3 = 38$$

Take the initial approximation as $X^{(0)} = (1, 1, 0)$
and do three iterations. (6.5)

4. (a) Obtain the piecewise linear interpolating polynomials for the function $f(x)$ defined by the data:

x	1	2	4	8
f(x)	3	7	21	73

(6)

(b) Calculate the Newton second order divided

difference $\frac{1}{x^2}$ of based on the points x_0, x_1, x_2 .

(6)

(c) Obtain the Lagrange form of the interpolating polynomial for the following data:

x	1	2	5
f(x)	-11	-23	1

(6)

5. (a) Find the highest degree of the polynomial for which the second order backward difference approximation for the first derivative

$$f'(x_0) \approx \frac{3f(x_0) - 4f(x_0 - h) + f(x_0 - 2h)}{2h}$$

provides the exact value of the derivative irrespective of h . (6)

- (b) Derive second-order forward difference approximation to the first derivative of a function f given by

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h}$$

(6)

- (c) Approximate the derivative of $f(x) = \sin x$ at $x_0 = \pi$ using the second order central difference formula taking $h = 1/2, 1/4$ and $1/8$ and then extrapolate from these values using Richardson extrapolation. (6)

6. (a) Using the Simpson's rule, approximate the value, of the integral $\int_2^5 \ln x \, dx$. Verify that the theoretical error bound holds. (6.5)

- (b) Apply Euler's method to approximate the solution of initial value problem $\frac{dx}{dt} = \frac{e^t}{x}, 0 \leq t \leq 2, x(0) = 1$ and $N = 4$.

Given that the exact solution is $x(t) = \sqrt{2e^t - 1}$, compute the absolute error at each step. (6.5)

(c) Apply the optimal RK2 method to approximate the solution of the initial value problem $\frac{dx}{dt} = 1 + \frac{x}{t}$,

$1 \leq t \leq 2$, $x(1) = 1$ taking the step size as $h = 0.5$.

(6.5)

Name of the Course: **LOCF B.Sc. (H) Mathematics**
 Unique Paper Code: **32357501**
 Name of the Paper: **DSE-1 (i) Numerical Analysis**
 Semester: **V**
 Duration: **3 hours**
 Maximum Marks: **75 Marks**

Attempt any four questions. All questions carry equal marks.

1. (a) Find the smallest positive root of the given equation by performing three iterations of the Bisection method

$$f(x) = e^x - 3x = 0.$$

- (b) Apply four iterations of the Fixed Point Iteration Method to find an approximate root of the following equation by taking the initial approximation as $p_0=0$

$$f(x) = 3x - (1 + \cos x) = 0.$$

Also represent the root graphically.

- (c) Find LU decomposition for the matrix

$$A = \begin{pmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & -2 & -1 \end{pmatrix}$$

and then solve the system

$$3x + y + 2z = 4$$

$$2x - 3y - z = -6$$

$$x - 2y - z = -4$$

2. (a) Set up the Gauss-Seidel iteration scheme for the following system of equations

$$2x - y + 2z = 3$$

$$x + 3y + 3z = -1$$

$$x + 2y + 5z = 1$$

and iterate three times starting with the initial vector $X^{(0)} = (0, 0, 0.5)^T$.

- (b) Apply Secant method to find a root of the equation

$$\ln(1+x) - \frac{1}{2} \cos x = 0$$

(0,1). Perform three iterations. What order of convergence do you expect?

- (c) Solve the following system of equations using SOR iteration method

$$5x_1 + x_2 - 2x_3 = 2$$

$$3x_1 + 4x_2 - x_3 = -2$$

$$2x_1 - 3x_2 + 5x_3 = 10$$

Take $w = 0.9$ with $X^{(0)} = (0,0,0)^T$ and iterate three times.

3. (a) Find a polynomial of degree 3 or less passing through the points $(-1,9)$, $(0,5)$, $(1,3)$ and $(2,1)$ using Lagrange interpolation. Use this polynomial to estimate the ordinate for $x = 1.5$.
 (b) Use central difference formula to approximate $f'(0.5)$ for the function $f(x) = 4e^{-2x}$ by taking two step sizes $h = 0.1$ and 0.05 . What is the order of approximation?

4. (a) Derive the following approximation of function $f'(x_0)$ for an arbitrary function $f(x)$:

$$f'(x_0) = \frac{-3f_0 + 4f_1 - f_2}{2h}$$

Hence show that the above approximation provides exact value of the first order derivatives for $f(x) = 1, x$ and x^2 but not for $f(x) = x^3$.

(b) Determine the step size h in an equidistant table for $f(x) = \sin x$ in $[0, \pi/4]$, if the error in magnitude in quadratic interpolation is less than or equal to 5×10^{-8} in magnitude.

(c) Obtain the piecewise linear interpolating polynomials for the function $f(x)$ defined by the given data.

x	0.5	1.5	2.5
$f(x)$	0.125	3.375	15.625

Interpolate at $x = 1.0$ and 2.0 .

5. (a) Approximate the value of the given integral by Simpson's 1/3 rule and the Trapezoidal rule

$$\int_0^3 \frac{1}{4+x^2} dx.$$

Calculate the difference between the actual value and the approximate value. Also find the error term for both.

(b) Apply the Newton Raphson Method to find a root of the equation

$$f(x) = x^3 - x^2 - 10x + 6 = 0. \text{ Perform three iterations with } x_0 = 1.$$

- 6 (a) Construct the Richardson extrapolation table to find the derivative of the function $f(x) = \tan^{-1} x$ at $x_0 = 3$ using the first-order forward difference approximation

$$D_h^{(1)} = \frac{f(x_o + h) - f(x_o)}{h}$$

taking $h=1, 0.5, 0.25, 0.125$.

(b) Apply the Modified Euler method to find approximate solution of the following initial value problem with four numbers of steps

$$\frac{dy}{dx} = xy, \quad (1 \leq x \leq 2), \quad y(1) = 1.$$

Also compare with the exact solution.

Name of Course : **B.A. (Prog.)**
 Unique Paper Code : **62357602**
 Name of Paper : **DSE: Numerical Analysis**
 Semester : **VI**
 Duration : **3 hours**
 Maximum Marks : **75 Marks**

Attempt any four questions. All questions carry equal marks.

1. Find an interval of unit length which contains the smallest positive root of the equation $f(x) = x^4 - x - 10 = 0$. Taking the end points of this interval as initial approximations, do two iterations each of the Secant method and Regula Falsi Method.

2. Find the number of significant digits for the following:

2410, 2.41, 0.00241, 2.410×10^4 , 2.4100×10^4 , 2.41000×10^6 .

If $x = 0.278143 \times 10^4$ and $y = 0.278456 \times 10^4$, find the number of significant digits in $x + y$, $x - y$, xy .

3. Given the following system of equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 4x_1 + 3x_2 - x_3 &= 6 \\ 3x_1 + 5x_2 + 3x_3 &= 4 \end{aligned}$$

- (i) Find the solution using the Gauss- Jordan method.
 (ii) Perform two iterations of the Gauss-Seidel method starting with $X^{(0)} = (1, 1, 1)$.

4. Generate the forward and backward difference table for the data

x	0	0.2	0.4	0.6	0.8
f(x)	0.12	0.46	0.74	0.9	1.2

Hence interpolate the values of $f(0.1)$ and $f(0.7)$ by using Gregory Newton forward and backward differences Interpolation formulae respectively.

5. Approximate the second order derivative of $f(x) = e^x$ at $x_0 = 0$, taking $h = 1, 0.1, 0.01$ by using the formula

$$f''(x) \approx \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2}$$

Also approximate the derivative of $f(x) = 1 + x + x^2 + x^3$ at $x_0 = 0$, taking $h = 1, 0.1$, and 0.01 by using the formula

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}$$

Find the order of approximations in both the cases.

6. Find the approximate value of $I = \int_0^1 \frac{dx}{1+x^3}$ using the Trapezoidal rule with 2, 4 and 8 equal subintervals. Improve the result by Romberg integration.

This question paper contains 7 printed pages]

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S. No. of Question Paper : 347

Unique Paper Code : 235651

G

Name of the Paper : Numerical Analysis and Statistics

Name of the Course : B.A. (Prog.) Discipline Course

Semester : VI

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

This question paper has six questions in all.

Attempt any two parts from each question.

All questions are compulsory.

Use of scientific calculator is allowed.

Candidate can ask for log/statistical table.

1. (a) (i) Perform three iterations by Bisection method to obtain the smallest positive root of the equation :

$$f(x) = x^3 - x - 4 = 0.$$

P.T.O.

- (ii) If a root of $f(x) = 0$ lies in the interval (a, b) , then what is minimum number of iterations required when the permissible error is ϵ . 6

- (b) A real root of the equation :

$$f(x) = x^3 - 5x + 1 = 0.$$

lies in the interval $(0, 1)$. Perform four iterations of Secant method to obtain this root. 6

- (c) Perform five iterations by Newton-Raphson method to find the root of $N^{1/2}$, where $N = 17$. Take initial approximation $x_0 = 3$. 6

2. (a) Consider the system of equations :

$$\begin{bmatrix} 1 & -a \\ -a & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

where a is a real constant. For which values of a the Gauss-Seidel method converges. 6

(b) Solve the following system of equations :

$$2x_1 + x_2 + x_3 - 2x_4 = -10$$

$$4x_1 + 0x_2 + 2x_3 + x_4 = 8$$

$$3x_1 + 2x_2 + 2x_3 + 0x_4 = 7$$

$$x_1 + 3x_2 + 2x_3 - x_4 = -5$$

Using the Gauss-elimination method with partial-pivoting.

6

(c) For the following system of equations :

$$-3x_1 + x_2 + 0x_3 = -2$$

$$2x_1 - 3x_2 + x_3 = 0$$

$$0x_1 + 2x_2 - 3x_3 = -1$$

(i) Show that Jacobi iteration scheme converges.

(ii) Starting with $X^0 = [0, 0, 0]^T$, iterate three times.

6

P.T.O.

3. (a) Find the unique polynomial of degree 2 or less such that :

$$f(1) = 1, f(3) = 27, f(4) = 64,$$

using Lagrange interpolating formula. Estimate $f(2)$. $6\frac{1}{2}$

- (b) For the following data :

$$f(0) = 1, f(1) = 14, f(2) = 15,$$

$$f(4) = 5, f(5) = 6, f(6) = 19$$

Obtain the polynomial using Newton divided difference interpolation. Estimate $f(3)$. $6\frac{1}{2}$

- (c) If

$$f(x) = 1/x,$$

find the divided difference $f[x_1, x_2, x_3, x_4]$. $6\frac{1}{2}$

4. (a) Calculate the coefficient of correlation from the following observations : 6

X	Y
2.52	550
2.49	610

2.47	730
2.42	870
1.69	880
3.43	930
4.72	400

- (b) Determine the line of regression of Y on X for the following data :

6

X	Y
65	67
66	68
67	65
67	68
68	72
69	72
70	69
72	71

- (c) Let the pmf $p(x)$ be positive at $x = -1, 0, 1$ and zero elsewhere. If

$$p(0) = \frac{1}{4} \text{ and if } E(X) = \frac{1}{4},$$

determine $p(-1)$ and $p(1)$.

6

5. (a) Let X have the pdf

$$f(x) = \frac{1}{x^2}, \quad 1 < x < \infty,$$

zero elsewhere. Show that $E(X)$ does not exist. $6\frac{1}{2}$

- (b) Let X be a random variable such that

$$E[(X - b)^2]$$

exists for all real b . Show that

$$E[(X - b)^2]$$

is minimum when $b = E(X)$.

 $6\frac{1}{2}$

- (c) Determine the mode of normal distribution. $6\frac{1}{2}$

6. (a) In a book of 520 pages, 390 typo-graphical errors occur. Assuming Poisson law for the number of error per page,

find the probability that a random sample of pages will contain no error.

(Given, $e^{-0.75} = 0.473$).

6½

(b) If X is a normal variate with mean 30 and S.D. 5. Find the probabilities that :

6½

(i) $26 \leq X \leq 40$

(ii) $X \geq 45$.

6½

(c) Determine the moment generating function of Binomial distribution.

6½

[This question paper contains 7 printed pages]

Your Roll No. :

Sl. No. of Q. Paper : **9650** **HC**

Unique Paper Code : 62357602

Name of the Course : **B.A.(Programme)**
Mathematics : DSE-4

Name of the Paper : Numerical Analysis

Semester : VI

Time : 3 Hours **Maximum Marks : 75**

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) All **six** questions are compulsory.
- (c) Attempt any **two** parts from each question.
- (d) Use of non-programmable scientific calculator is allowed.

1. (a) Perform three iterations of Secant method to obtain a root of the equation 6

$$x^2 - 7 = 0$$

with initial approximation $x_0 = 2, x_1 = 3$.

P.T.O.

- (b) Perform four iterations of bisection method to obtain the root of the equation

$$x^3 - 3x^2 - 0.09x + 0.27 = 0 \quad 6$$

in the interval $[0,1]$.

- (c) If true value is 0.000310698 and approximated value is 0.0049065, then find the absolute and the relative error. Differentiate between round-off error and truncation error. 6 3

2. (a) Perform two iterations of Newton's method to solve the non-linear system of equations 6

$$x^2y + y^3 = 10$$

$$xy^2 - x^2 = 3$$

with initial approximation $(x_0, y_0) = (0.8, 2.2)$.

- (b) Perform three iterations of Regula Falsi method to find the root of the equation

$$f(x) = x^3 - 2 = 0 \quad 6$$

in the interval $[1,2]$.

- (c) Perform three iterations of Newton-Raphson method to obtain the cube root of 17 with initial approximation $x_0 = 2$.

6

3. (a) Find the inverse of the following matrix using the Gauss Jordan method :

6.5

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

- (b) Solve the tridiagonal system $AX = b$ by using Gauss- Thomas algorithm.

$$\begin{pmatrix} 1 & 2 & 0 \\ 1 & 3 & 3 \\ 0 & 3 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 10 \\ 17 \\ 22 \end{pmatrix}$$

6.5

(c) For the following system of equations:

6.5

$$10x + 4y - 2z = 12$$

$$x - 10y - z = -10$$

$$5x + 2y - 10z = -3$$

(i) Show that Jacobi iteration scheme converges.

(ii) Perform three iterations of Jacobi iteration scheme starting with initial vector $(x, y, z) = (0, 0, 0)$.

4. (a) For the following data, obtain the Gregory - Newton backward difference polynomial.

x	0.1	0.2	0.3	0.4	0.5
f(x)	1.40	1.56	1.76	2.00	2.28

Also find the approximate value of f at $x = 0.25$.

6

(b) Prove the following :

$$1 + \Delta = 1 + \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$$

(c) The function $f(x) = \sin(x)$ is defined on $[1, 3]$. Find Lagrange interpolating linear polynomial on intervals $[1, 2]$ and $[2, 3]$. Hence find the approximate value of $f(1.5)$ and $f(2.5)$. 6

5. (a) The following data for the function $f(x) = x^4$ is given (6.5)

X	0.4	0.6	0.8
f(x)	0.0256	0.1296	0.4096

Find $f'(0.8)$ and $f''(0.8)$ using quadratic interpolation.

(b) Find integral of $I = \int_0^2 e^{-x^2} dx$ using Simpson's rule and Trapezoidal rule. (6.5)

(c) Evaluate $\int_0^1 \frac{dx}{1+x}$ using composite Trapezoidal rule and Romberg integration with $h=1$ and $h=1/2$ only. 6.5

6. (a) Employ the classical fourth order RK method to integrate $y' = 4e^{0.8t} - 0.5y$, $y(0) = 2$ from $t = 0$ to 1 using a step size of 1. 6.5

(b) Apply Euler's method to approximate the solution of

$$y' = x + y, \quad y(0) = 2, \quad 6.5$$

and calculate $y(1)$ by using $h=0.5$.

- (c) Apply finite difference method (second order) to solve the given problem :

6.5

$$y'' = y + x, \quad y(0) = 0, \quad y(1) = 0,$$

with $h = 1/4$.



This question paper contains 4 printed pages.

Your Roll No.

S. No. of Paper : 3404

IC

Unique Paper Code : 62357602

Name of the Paper : Numerical Analysis

Name of the Course : B.A. (Prog.) Maths : DSE-2B

Semester : VI

Duration : 3 hours

Maximum Marks : 75

(Write your Roll No. on the top immediately
on receipt of this question paper.)

All six questions are compulsory. Attempt any two parts from
each question. Use of non-programmable simple calculator
is allowed.

1. (a) Define Rounding-off error with examples. If 0.333 is
the approximate value of $\frac{1}{3}$, find absolute and relative
errors. 6

- (b) Perform two iterations of Newton's method to solve
the non-linear system of equations:

$$y \cos(xy) + 1 = 0$$

$$\sin(xy) + x - y = 0$$

with initial approximation $(x_0, y_0) = (1, 2)$. 6

P. T. O.

- (c) Perform four iterations of bisection method to obtain the root of the equation:

$$x^3 - 3x^2 - 0.16x + 0.48 = 0$$

in the interval $[0,1]$.

6

2. (a) Perform three iterations of Regula Falsi method to find the root of the equation:

$$f(x) = x^2 - 5 = 0$$

in the interval $[2,3]$.

6

- (b) Perform three iterations of Newton-Raphson method to find the square root of $3/4$ with initial approximation $x_0 = 0.5$.

6

- (c) Perform three iterations of secant method to find the root of the equation:

$$x^3 - 9x + 2 = 0$$

in the interval $[0,1]$.

6

3. (a) Solve the following system of linear equations using the Gauss elimination method with partial pivoting:

$$x + 2y + 3z = 1$$

$$2x + 4y + 10z = -2$$

$$3x + 14y + 28z = -8$$

6.5

- (b) Perform three iterations of Gauss-Seidel iteration method for the following system of equations:

$2x - y = 7$, $-x + 2y - z = 1$, $-y + 2z = 1$,
starting with initial solution as $(x, y, z) = (0, 0, 0)$. 6.5

- (c) Explain Gauss-Thomas algorithm and solve the tri-diagonal system $AX = b$ by using Gauss-Thomas method.

$$\begin{pmatrix} 0.8 & -0.4 & 0 \\ -0.4 & 0.8 & -0.4 \\ 0 & -0.4 & 0.8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 41 \\ 25 \\ 105 \end{pmatrix} \quad 6.5$$

4. (a) Given that:

$$f(-2) = 4, f(0) = 2, f(2) = 8$$

Find the unique polynomial of degree 2 by Lagrange interpolation. Also find bound on the error. 6

- (b) For the following data, calculate the differences and obtain the Gregory-Newton forward difference polynomial:

x	0.1	0.2	0.3	0.4	0.5
$f(x)$	1.40	1.56	1.76	2.00	2.28

Also find the approximate value of $f(0.35)$. 6

- (c) (i) Find the maximum value of step size h that can be used to tabulate $f(x) = e^x$ on $[0, 1]$ using linear interpolation such that $|\text{error}| \leq 5 \times 10^{-4}$.

- (ii) Let $f(x) = \log_e(1+x)$, $x_0 = 1$ and $x_1 = 1.1$.

Approximate $f(1.04)$ by using Newton divided difference interpolating polynomial. 6

5.(a) Find the approximate value of $I = \int_0^4 2^x dx$ using (i) Trapezoidal rule, (ii) Simpson's rule. 6.5

(b) Find the approximate value of $I = \int_0^1 \frac{\sin x}{x} dx$ by using Newton Cote's open formula for three-point rule. 6.5

(c) Evaluate $\int_0^1 \left(1 + \frac{\sin x}{x}\right) dx$ using composite Trapezoidal rule and Romberg integration with $h=1$ and $h=1/2$ only. 6.5

Q6 (a) Apply Euler's modified method to approximate the solution of the following initial-value problem and calculate $y(2)$ by using $h=1$:

$$y' = 4e^{0.8y} - 0.5y, \quad y(0) = 2. \quad 6.5$$

(b) Employ the classical fourth order RK method to integrate $y' = -2ty^2$, $y(0) = 1$ with $h = 0.2$ on the interval $[0, 0.2]$. 6.5

(c) Apply finite difference method to solve the given problem:

$$\frac{d^2 y}{dx^2} = y + x(x-4), \quad 0 \leq x \leq 4$$

with $h=1$ and $y(0)=0$, $y(4)=0$. 6.5

This question paper contains 4 printed pages.

Your Roll No.

S. No. of Paper : 3376

IC

Unique Paper Code : 62357602

Name of the Paper : Numerical Analysis

Name of the Course : B.A. (Prog.) Maths.: DSE-1B

Semester : VI

Duration : 3 hours

Maximum Marks : 75

***(Write your Roll No. on the top immediately
on receipt of this question paper.)***

***All six questions are compulsory. Attempt any two parts from
each question. Use of simple calculator is allowed.***

1. (a) Perform three iterations of Newton-Raphson method to obtain root of the equation:

$$f(x) = \cos x - xe^x = 0$$

with initial approximation $x_0 = 1$.

6

- (b) Define Truncation Error. Evaluate the sum:

$$\sqrt{3} + \sqrt{5} + \sqrt{7}$$

to four significant digits and find its absolute and relative errors.

6

- (c) Perform three iterations of Regula Falsi method to obtain the root of the equation:

P. T. O.

$$f(x) = x^3 - 2x^2 - 5 = 0 \quad (6)$$

in the interval $[2, 3]$.

6

2. (a) Perform three iterations of secant method to obtain the square root of 3 with initial approximation:

$$x_0 = 1, x_1 = 2.$$

6

- (b) Perform four iterations of bisection method to find the root of the equation:

$$x^3 - 2x^2 - 0.04x + 0.08 = 0$$

in the interval $[0, 1]$.

6

- (c) Perform two iterations of Newton's method to solve the non-linear system of equations:

$$x^2 + xy + y^2 = 7$$

$$x^3 + y^3 = 9$$

with initial approximation $(x_0, y_0) = (1.5, 0.5)$.

6

3. (a) Solve the following system of linear equations using the Gauss elimination method with partial pivoting:

$$2x + 6y + 10z = 0$$

$$x + 3y + 3z = 2$$

$$3x + 14y + 28z = -8$$

6.5

- (b) Perform three iterations of Gauss-Seidel iteration method for the following system of equations:

$$5x + y - 2z = 2$$

$$3x + 4y - z = -2,$$

$$2x - 3y + 5z = 10,$$

starting with initial solution as $(x, y, z) = (0, 0, 0)$.

6.5

(c) Solve the following system of equations by using the Gauss-Jordan method:

$$\begin{pmatrix} 2 & 1 & 3 \\ 4 & -3 & 5 \\ -3 & 2 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -7 \\ -3 \end{pmatrix} \quad 6.5$$

4. (a) Prove the following relation:

$$\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2} \quad 6$$

(b) Given the following data:

x	0.1	0.2
$\sin(x)$	0.09998	0.1986

Find Lagrange interpolating polynomial and approximate the value $\sin(0.15)$. Obtain a bound on the truncation error also. 6

(c) The following data represents the function $f(x) = x^{\frac{1}{3}}$:

x	0	1	8
$f(x)$	0	1	2

Find Newton divided difference interpolating polynomial of degree 2. Also find the approximate value of $f(7)$ and compare with the exact value. 6

5. (a) Find $f''(2.0)$ using quadratic interpolation using the following data:

x	2.0	2.2	2.6
$f(x)$	0.69315	0.78846	0.95551

Obtain an upper bound on error also. 6.5

- (b) Find the approximate value of $I = \int_0^1 \frac{\sin x}{x} dx$, by using Newton's Cote open formula (i) mid- point rule (ii) two- point rule. 6.5

- (c) Evaluate the integral $I = \int_0^2 \frac{dx}{3+4x}$, using Gauss Quadrature 3- point rule. 6.5

6. (a) Use the Euler method to solve boundary value problem $y' = 4e^{0.8t} - 0.5y$, $y(0) = 2$ on the interval $[0,3]$ with $h = 1$. 6.5

- (b) Given the initial value problem:

$$y' = -2ty^2, \quad y(0) = 1$$

estimate $y(0.4)$ using Ralston's method (R. K. Method 2nd Order) with $h = 0.2$. 6.5

- (c) Using a second order finite difference method with $h = 1$, find the solution of the Boundary Value Problem $y'' - y = x(x-4)$, $0 \leq x \leq 4$ with $y(0) = y(4) = 0$. 6.5